

CHAPTER 6

SAMPLING AND ANALOG TO DIGITAL CONVERSION

- The Analog-to-digital (A/D) converter enables digital communication systems to convey continuous-analog signals such as audio and video.
- The A/D conversion takes place in three main steps :
 - **Sampling:** converts a continuous-analog signal into a discrete-analog signal.
 - **Quantization:** converts the discrete-analog signal into digital signal.
 - **Coding:** converts the digital signal into binary code.

6.1 The Sampling Process

- Let $g(t)$ be a continuous-analog time signal whose bandwidth is B Hz. That is, its spectrum extends from 0 to B Hz. Mathematically $G(f) = 0$ for $|f| > B$.
- Bandwidth limiting $g(t)$ is achieved by passing $g(t)$ through a low pass filter with a cutoff frequency $= B$ Hz. This filter is called the anti-aliasing filter because it avoids aliasing between the spectrums of the sampled signal.
- Sampling $g(t)$ at a rate of f_s Hz means that we take f_s uniform samples per second from $g(t)$. f_s is called the sampling frequency or sampling rate measured by samples/second.
- Uniform sampling can be accomplished by multiplying $g(t)$ by an impulse train $\delta_{T_s}(t)$ consisting of unit impulses repeating periodically every T_s seconds, where $T_s = 1/f_s$. T_s is called the sampling interval.

- Practically, the sampling process can be implemented by passing $g(t)$ through an electronic switch with switching speed equal to f_s as shown in Figure (6.1).

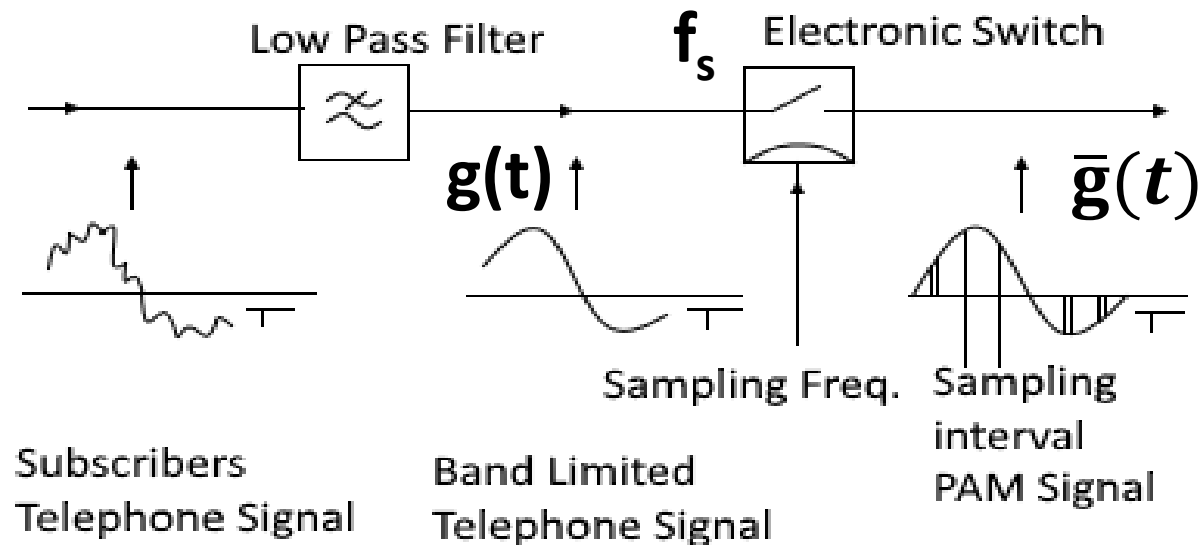


Figure (6.1)

- The sampling theorem** states that if the highest frequency in the signal spectrum is B Hz, and it is sampled with sample rate f_s samples per second, then the signal can be reconstructed from its discrete samples if $f_s \geq 2B$.

- This means that to preserve the information from a continuous-time signal, we need transmit only its samples.
- The sampled signal consists of impulses spaced every T_s seconds (the sampling interval). The n_{th} impulse, located at $t = nT_s$, has a strength $g(nT_s)$ which is the value of $g(t)$ at $t = nT_s$ as shown in Figure (6.2 a, c and d).
- Thus, the relationship between the sampled signal $\bar{g}(t)$ and the original analog signal $g(t)$ is given by:

$$\bar{g}(t) = g(t)\delta_{T_s}(t) = \sum_n g(nT_s)\delta(t - nT_s) \quad (6.1)$$

- Because the impulse train $\delta_{T_s}(t)$ is a periodic signal of period T_s , it can be expressed as an exponential Fourier series, already found in Example 3. 11 as

$$\delta_{T_s}(t) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} e^{jn\omega_s t} \quad \omega_s = \frac{2\pi}{T_s} = 2\pi f_s \quad (6.2)$$

Therefore,

$$\bar{g}(t) = g(t)\delta_{T_s}(t) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} g(t)e^{jn2\pi f_s t} \quad (6.3)$$

- To find $\bar{G}(f)$, we take the Fourier transform of the summation in Eq. (6.3). Based on the frequency-shifting property, the transform of the n_{th} term is shifted by nf_s , therefore

$$\bar{G}(f) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} G(f - nf_s) \quad (6.4)$$

- This means that the spectrum $\bar{G}(f)$ consists of $G(f)$, scaled by a constant $1 / T_s$, repeating periodically with period $f_s = 1 / T_s$ Hz, as shown in Fig. 6.1e.
- Graphically from Fig. 6.1e , perfect recovery of the original signal $g(t)$ is possible if there is no overlap among the replicas $\bar{G}(f)$.
- This requires $f_s > 2B$ (6.5)
- When $f_s = 2B$, f_s is called the **Nyquist rate**.
- Also, the sampling interval $T_s = 1/f_s$. Therefore,

$$T_s < \frac{1}{2B} \quad (6.6)$$

- The sampling interval $T_s = 1 / 2B$ is called the **Nyquist interval** for the low-pass signal $g(t)$.

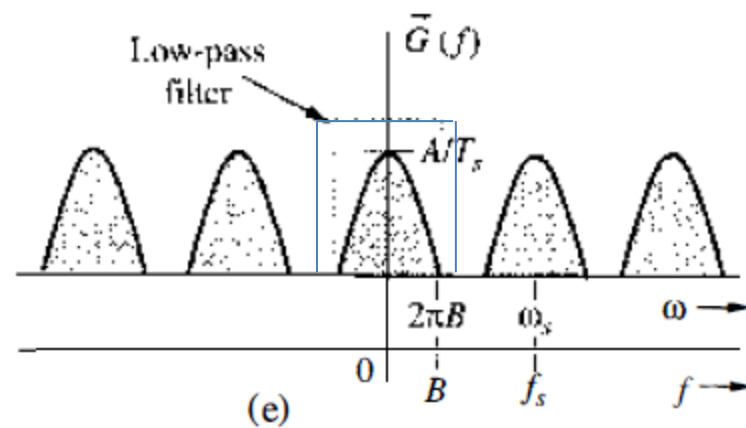
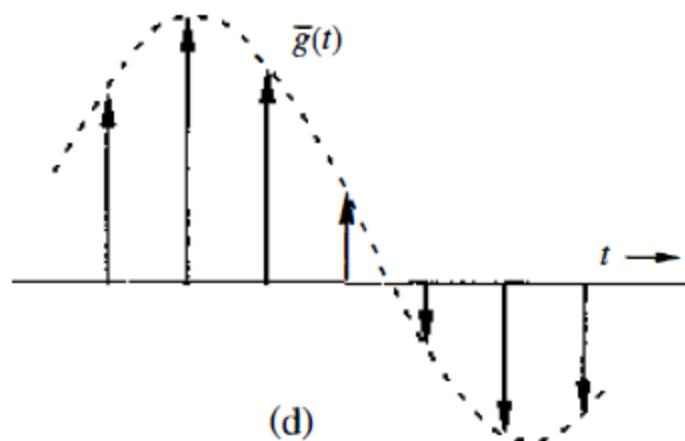
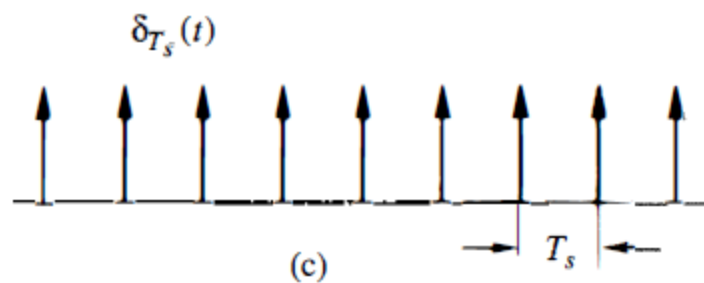
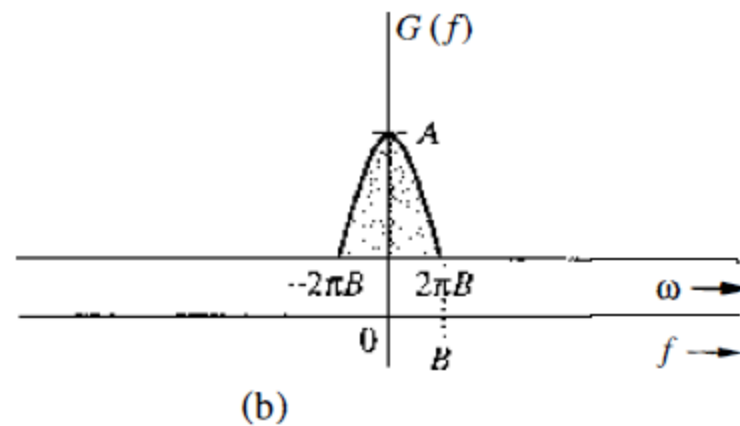
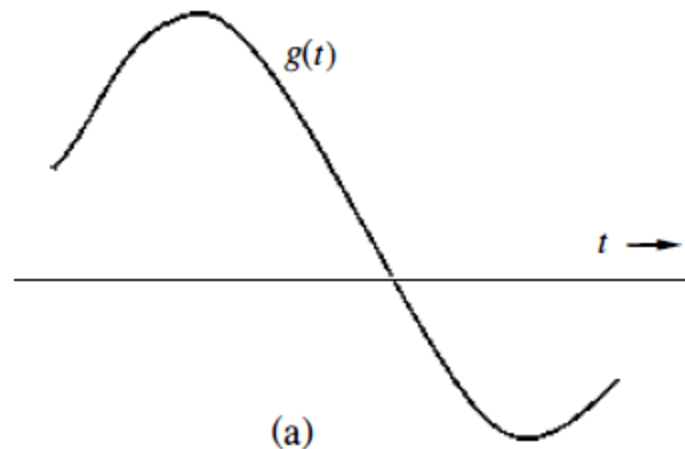


Figure (6.2)

- If $f_s < 2B$, the spectra of the replica overlap leading to aliasing error as shown in Figure (6.3).

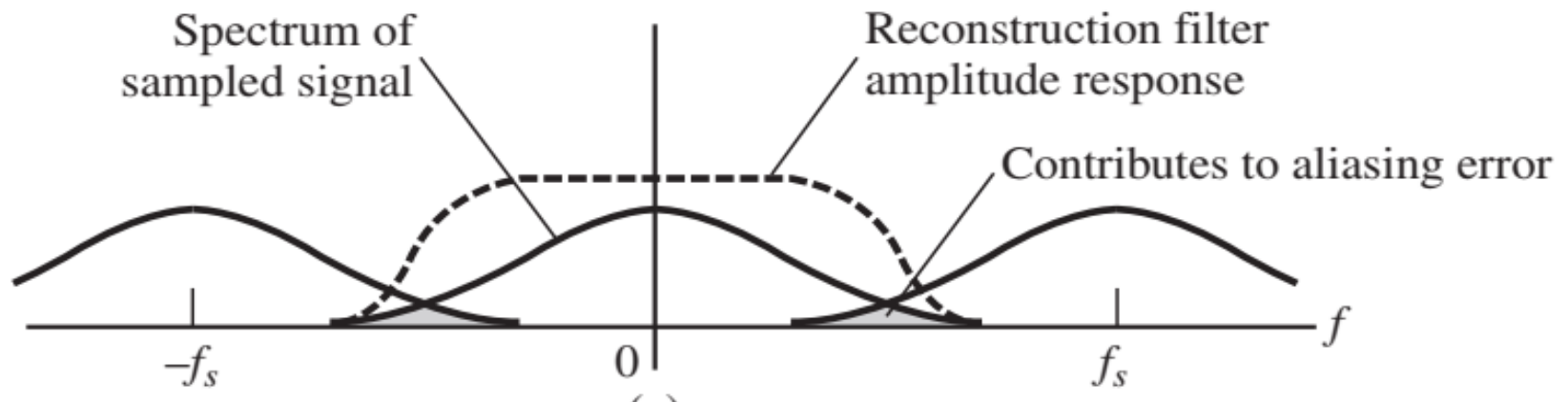


Figure (6.3)

- To avoid aliasing, the signal $g(t)$ is passed through a low pass filter **before sampling**. As mentioned previously, this filter is called the antialiasing filter.

Practical Reconstruction Filters

- If a signal is sampled at the Nyquist rate $f_s = 2B$ Hz, the spectrum $\bar{G}(f)$ consists of repetitions of $G(f)$ without any gap between successive cycles, as shown in Fig. 6.4a. To recover $g(t)$ from $\bar{g}(t)$, we need to pass the sampled signal $\bar{g}(t)$ through an ideal low-pass filter with sharp cutoff characteristic (dotted area in Fig. 6.4a). Such a filter is unrealizable in practice.
- A practical solution to this problem is to sample the signal at a rate higher than the Nyquist rate ($f_s > 2B$ or $\omega_s > 4\pi B$). This yields $\bar{G}(f)$, consisting of repetitions of $G(f)$ with a finite band gap between successive cycles, as shown in Fig. 6.4b. We can now recover $G(f)$ from $\bar{G}(f)$ by using a realizable low-pass filter with a gradual cutoff characteristic (dotted area in Fig. 6.4b).

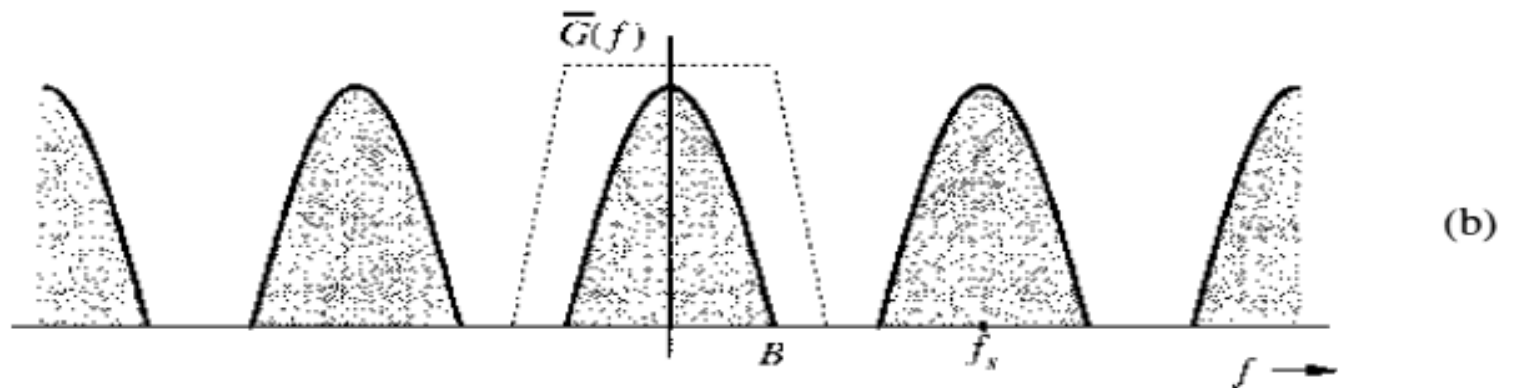
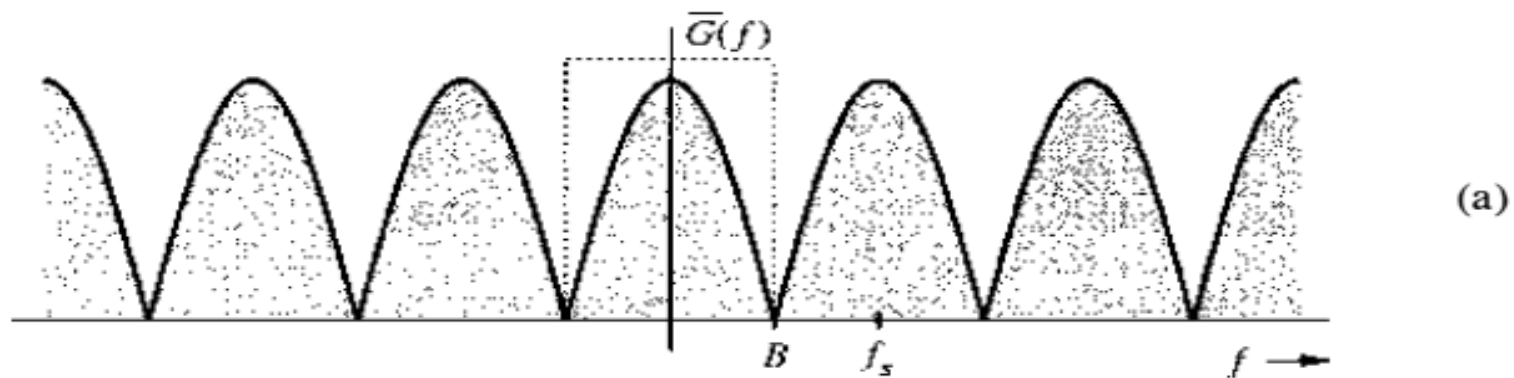


Figure (6.4)

Some Applications of the Sampling Theorem

➤ Pulse Modulation (PM)

- By pulse modulation, the sample values are used to modify certain parameters of a periodic pulse train as follows:
- Pulse amplitude modulation (**PAM**): the amplitudes of the pulses is varied in proportion to the sample values of the signal $g(t)$ (Fig. 6.5b).
- Pulse width modulation (**PWM**): the widths of the pulses is varied in proportion to the sample values of the signal $g(t)$ (Fig. 6.5c).
- Pulse position modulation (**PPM**): the positions of the pulses is varied in proportion to the sample values of the signal $g(t)$ (Fig. 6.5d).

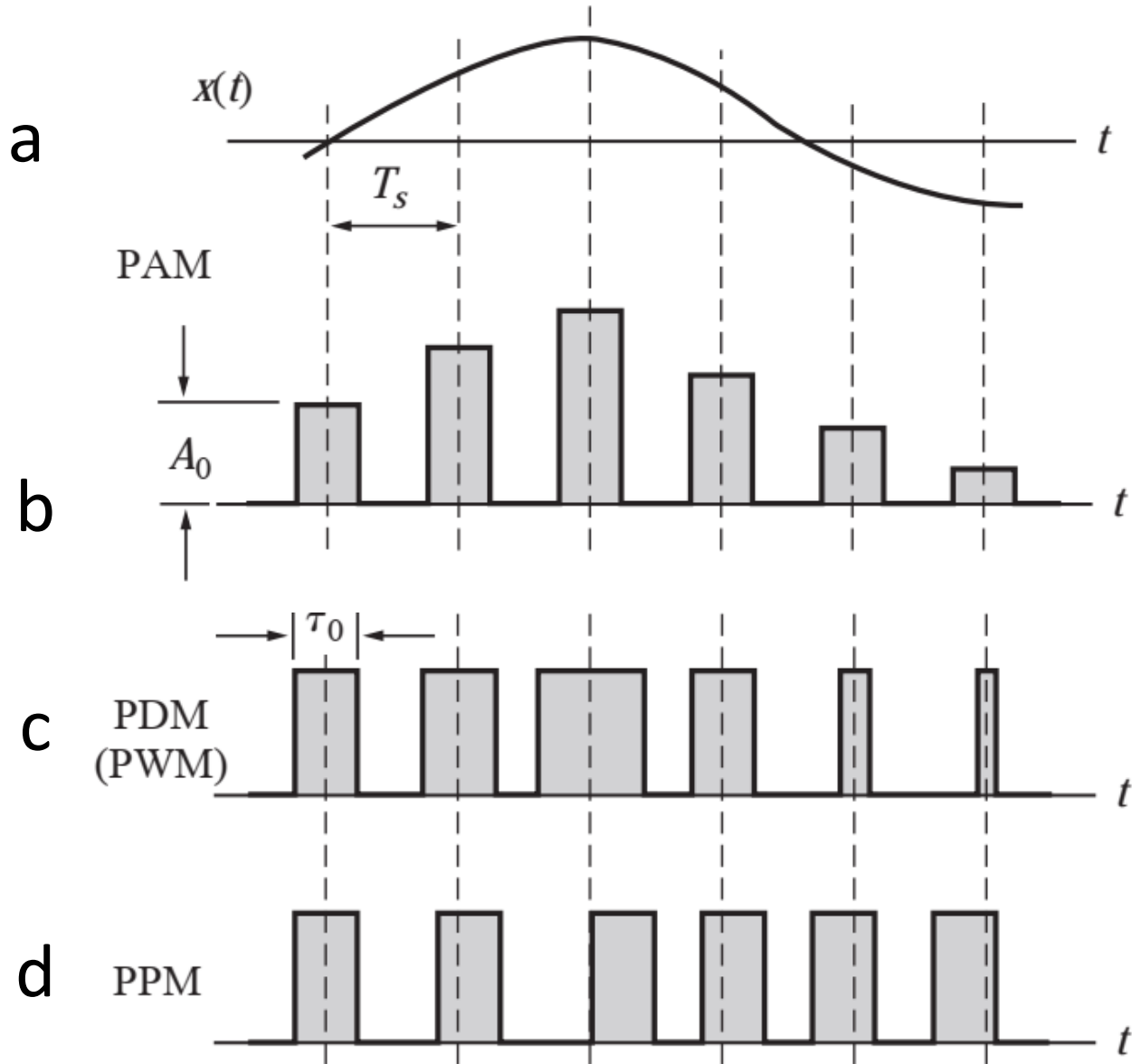


Figure (6.5)

➤ Time Division Multiplexing (TDM)

Pulse modulation permits the simultaneous transmission of several signals on a time-sharing basis-time **division multiplexing (TDM)**. Because a pulse-modulated signal occupies only a part of the channel time, we can transmit several pulse modulated signals on the same channel by interweaving them. Figure (6.6) shows the TDM of two PAM signals. In this manner we can multiplex several signals on the same channel by reducing pulse widths

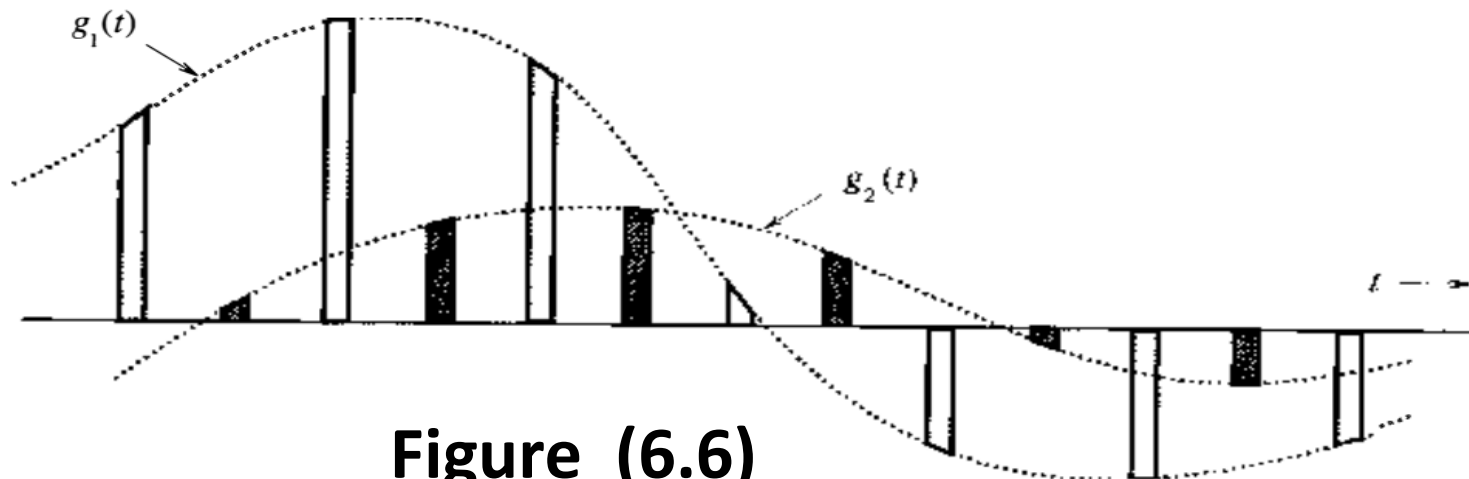


Figure (6.6)